```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import (
    preprocess_input, decode_predictions
)

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)

if image is None:
    raise ValueError("Unable to read the uploaded image.")

# Display input image
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Input Image")
plt.axis("off")
plt.show()

# Load pretrained model
model = MobileNetV2(weights="imagenet")

# Preprocess image
img = cv2.resize(image, (224, 224))
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
img = np.expand_dims(img, axis=0)
img = preprocess_input(img)

# Predict image category
predictions = model.predict(img, verbose=0)
results = decode_predictions(predictions, top=5)[0]

print("Top 5 Predictions:")
for _, label, confidence in results:
    print(label, ":", round(confidence * 100, 2), "%")

# Display best prediction
best_label = results[0][1]
confidence = results[0][2] * 100

plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title(f"Detected: {best_label} ({confidence:.2f}%)")
plt.axis("off")
plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823 (3).png
Image uploaded successfully: Screenshot 2026-10-10 075823 (3).png
```




```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/mobilenet_v2/mobilene
14536120/14536120 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Downloading data from https://storage.googleapis.com/download.tensorflow.org/data/imagenet_class_index.j
35363/35363 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Top 5 Predictions:
lion : 99.73 %
hartebeest : 0.01 %
ox : 0.01 %
bull_mastiff : 0.01 %
leopard : 0.01 %
```



Welcome to Colab!